Economia das Organizações e dos Incentivos — MPE Insper

Exercise List: Real Book Exercises, Lecture 2

How to use this list. Every exercise below is a real exercise from one of the course books, reproduced verbatim: the source line gives author, book, chapter, exercise, the printed page and, in brackets, the page your PDF viewer shows for the file in `Livros/`, and whether the source gives the answer. Muñoz-Garcia & Toro-Gonzalez print a worked answer right after each exercise; MWG answers are in the MWG solutions manual in `Livros/`; Tadelis, Gibbons and Cabral give no answer in the book. When an exercise refers to a model, game or figure stated elsewhere in its book, that material is copied into the same block under "Referenced material", so every exercise can be solved where it is printed. There is one block per lecture of the real schedule. Each block has ten exercises in strictly rising difficulty, rated from 1 (apply a definition) to 5 (graduate level) next to each heading: three foundations, five exam level, two advanced. Try each exercise in the space provided before opening the book's answer.

Difficulty path. Lecture 2: from writing a matrix (1) through the Bertrand uniqueness proof and MWG 8.D.2 to MWG 8.D.6 (5), a proof that every 2×2 game, with arbitrary payoffs, has a mixed-strategy equilibrium.

Block	#	Source	p. [PDF]	Level	What it trains	Diff.	Answer
L2	1 (11)	Tadelis 5.6	96 [115]	Found.	Hawk-dove: matrix, pure equilibria	1	none
	2 (12)	Muñoz-Garcia Ch. 2 Ex. 2‡	41 [54]	Found.	Battle of the sexes: pure equilibria	1	book
	3 (13)	Muñoz-Garcia Ch. 3 Ex. 1	86–87 [99–100]	Found.	Chicken: pure and mixed equilibria	2	book
	4 (14)	Gibbons 1.12‡	51 [59]	Exam	A mixed equilibrium by indifference	2	none
	5 (15)	Tadelis 5.8	96–97 [115–116]	Exam	Cournot with n firms; competitive limit	3	none
	6 (16)	Gibbons 1.6	49 [57]	Exam	Cournot, asymmetric costs; corner case	3	none
	7 (17)	Gibbons 1.7‡	49–50 [57–58]	Exam	Homogeneous Bertrand: $p_1 = p_2 = c$ is the unique equilibrium	3	none
	8 (18)	Tadelis 6.8	125 [144]	Exam	Three-player entry; symmetric mixing	4	none
	9 (19)	MWG 8.D.2‡	263 [142]	Adv.	Proof: a unique IESDS survivor is a Nash equilibrium	4	manual
	10 (20)	MWG 8.D.6	264 [143]	Adv.	A mixed equilibrium exists in every 2×2 game	5	manual

#: item number in this lecture's list, then its number in the combined list (`lista-reais-lectures-01-05.pdf`) in brackets. Items not in the list for Lectures 1–4 (`lista-reais-lectures-01-04.tex`) are marked: † added in the 30-item version, ‡ added in the 50-item version. Every other item is reused from that list with its statement unchanged, some in a different block or position. "Diff.": difficulty from 1 (apply a definition) to 5 (graduate level). Pages: printed page, then the PDF viewer page in brackets (offsets in `Livros/book-registry.json`; MWG is a two-pages-per-sheet scan). "Answer": *book* = worked answer printed after the exercise; *manual* = MWG solutions manual in `Livros/`; *none* = no answer in the book. Books: Tadelis (2013), *Game Theory: An Introduction*; Gibbons (1992), *Game Theory for Applied Economists*; Muñoz-Garcia & Toro-Gonzalez (2019), *Strategy and Game Theory: Practice Exercises with Answers*, 2nd ed.; Mas-Colell, Whinston & Green (1995), *Microeconomic Theory* (MWG); Cabral (2017), *Introduction to Industrial Organization*, 2nd ed.

Block 2 — Static games II: Nash equilibrium and mixed strategies *(Lecture 2)*

Exercise 1. (item 11 of the combined list) *Hawk-Dove* [Foundations; difficulty 1/5]

Source. Tadelis (2013), *Game Theory: An Introduction*, Ch. 5 "Pinning Down Beliefs: Nash Equilibrium", Exercise 5.6 "Hawk-Dove", p. 96 [PDF 115]. No answer in the book. Reproduced verbatim, with the book's footnote 12.

5.6 Hawk-Dove: The following game has been widely used in evolutionary biology to understand how fighting and display strategies by animals could coexist in a population. For a typical Hawk-Dove game there are resources to be gained (e.g., food, mates, territories), denoted as v. Each of two players can choose to be aggressive, as Hawk (H), or compromising, as Dove (D). If both players choose H then they split the resources but lose some payoff from injuries, denoted as k. Assume that $k > \frac{v}{2}$. If both choose D then they split the resources but engage in some display of power that carries a display cost d, with $d < \frac{v}{2}$. Finally, if player i chooses H while j chooses D then i gets all the resources while j leaves with no benefits and no costs.

a. Describe this game in a matrix.

b. Assume that $v = 10$, $k = 6$, and $d = 4$. What outcomes can be supported as pure-strategy Nash equilibria?[12]

[12] In the evolutionary biology literature, the analysis performed is of a very different nature. Instead of considering the Nash equilibrium analysis of a static game, the analysis is a dynamic one in which successful strategies "replicate" in a large population. This analysis is part of a methodology called evolutionary game theory. For more on the subject see Gintis (2000).

Resolution:

Exercise 2. (item 12 of the combined list) *Battle of the Sexes* [Foundations; difficulty 1/5]

Source. Muñoz-Garcia & Toro-Gonzalez (2019), *Strategy and Game Theory: Practice Exercises with Answers*, 2nd ed., Ch. 2 "Pure Strategy Nash Equilibrium and Simultaneous-Move Games with Complete Information", Exercise 2 "Battle of the Sexes"[A], p. 41 [PDF 54]. Worked answer in the book, right after the statement. Reproduced verbatim (statement only); Fig. 2.4, printed inside the answer on the same page, rebuilt, with the book's row label "Housband" kept as printed.

A husband and a wife are leaving work, and do not remember which event they are attending to tonight. Both of them, however, remember that last night's argument was about either attending to the football game (the most preferred event for the husband) or the opera (the most preferred event for the wife). To make matters worse, their cell phones broke, so they cannot call each other to confirm which event they are attending to. As a consequence, they must simultaneously and independently decide whether to attend to the football game or the opera.

The payoff matrix in Fig. 2.4 describes the preference of the husband (wife) for the football game (opera, respectively), but also indicates that both players prefer to be together rather than being alone (even if they are alone at their most preferred event). Find the set of Nash Equilibria of this game.

		Wife	
		Football	*Opera*
Housband	*Football*	3, 1	0, 0
	Opera	0, 0	1, 3

Fig. 2.4 Battle of the sexes game (Normal-form representation)

Resolution:

Exercise 3. (item 13 of the combined list) *Game of Chicken* [Foundations; difficulty 2/5]

Source. Muñoz-Garcia & Toro-Gonzalez (2019), *Strategy and Game Theory: Practice Exercises with Answers*, 2nd ed., Ch. 3 "Mixed Strategies, Strictly Competitive Games, and Correlated Equilibria", Exercise 1 "Game of Chicken"[A], pp. 86–87 [PDF 99–100]. Worked answer in the book, right after the statement. Reproduced verbatim (statement only).

Consider the game of Chicken (depicted in Fig. 3.1), in which two players driving their cars against each other must decide whether or not to swerve.

Part (a) Is there any strictly dominated strategy for Player 1? And for Player 2?

Part (b) What are the best responses for Player 1? And for Player 2?

Part (c) Can you find any pure strategy Nash Equilibrium (psNE) in this game?

Part (d) Find the mixed strategy Nash Equilibrium (msNE) of the game. *Hint*: denote by p the probability that Player 1 chooses Straight and by $(1-p)$ the probability that he chooses to Swerve. Similarly, let q denote the probability that Player 2 chooses Straight and $(1-q)$ the probability that she chooses to Swerve.

Part (e) Show your result of part (d) by graphically representing every player i's best response function $BRF_i(s_j)$, where $s_j = \{Swerve, Straight\}$ is the strategy selected by player $j \neq i$.

		Player 2	
		Straight	Swerve
Player 1	Straight	0,0	3,1
	Swerve	1,3	2,2

Fig. 3.1 Game of Chicken

Resolution:

Exercise 4. (item 14 of the combined list) *A mixed equilibrium by indifference*

[Exam level; difficulty 2/5]

Source. Gibbons (1992), *Game Theory for Applied Economists*, Ch. 1 "Static Games of Complete Information", §1.5 Problems (Section 1.3), Problem 1.12, p. 51 [PDF 59]. No answer in the book. Reproduced verbatim.

1.12. Find the mixed-strategy Nash equilibrium of the following normal-form game.

	L	R
T	2, 1	0, 2
B	1, 2	3, 0

Resolution:

Exercise 5. (item 15 of the combined list) *The n-Firm Cournot Model* [Exam level; difficulty 3/5]

Source. Tadelis (2013), *Game Theory: An Introduction*, Ch. 5 "Pinning Down Beliefs: Nash Equilibrium", Exercise 5.8 "The n-Firm Cournot Model", pp. 96–97 [PDF 115–116]. No answer in the book. Reproduced verbatim (the printed "$Q = q_i + \cdots + q_n$" is kept as printed).

5.8 The n-Firm Cournot Model: Suppose there are n firms in the Cournot oligopoly model. Let q_i denote the quantity produced by firm i, and let $Q = q_i + \cdots + q_n$ denote the aggregate production. Let $P(Q)$ denote the market clearing price (when demand equals Q) and assume that the inverse demand function is given by $P(Q) = a - Q$, where $Q \leq a$. Assume that firms have no fixed cost and that the cost of producing quantity q_i is cq_i (all firms have the same marginal cost, and assume that $c < a$).

a. Model this as a normal-form game.

b. What is the Nash (Cournot) equilibrium of the game in which firms choose their quantities simultaneously?

c. What happens to the equilibrium price as n approaches infinity? Is this familiar?

Resolution:

Exercise 6. (item 16 of the combined list) *Cournot with asymmetric costs* [Exam level; difficulty 3/5]

Source. Gibbons (1992), *Game Theory for Applied Economists*, Ch. 1 "Static Games of Complete Information", §1.5 Problems (Section 1.2), Problem 1.6, p. 49 [PDF 57]. No answer in the book. Reproduced verbatim.

1.6. Consider the Cournot duopoly model where inverse demand is $P(Q) = a - Q$ but firms have asymmetric marginal costs: c_1 for firm 1 and c_2 for firm 2. What is the Nash equilibrium if $0 < c_i < a/2$ for each firm? What if $c_1 < c_2 < a$ but $2c_2 > a + c_1$?

Referenced material. Source. "the Cournot duopoly model": Gibbons (1992), §1.2.A "Cournot Model of Duopoly", p. 15 [PDF 23], the paragraph that sets up the model. Reproduced verbatim; footnote 6 (a pointer to later sections) omitted.

Let q_1 and q_2 denote the quantities (of a homogeneous product) produced by firms 1 and 2, respectively. Let $P(Q) = a - Q$ be the market-clearing price when the aggregate quantity on the market is $Q = q_1 + q_2$. (More precisely, $P(Q) = a - Q$ for $Q < a$, and $P(Q) = 0$ for $Q \geq a$.) Assume that the total cost to firm i of producing quantity q_i is $C_i(q_i) = cq_i$. That is, there are no fixed costs and the marginal cost is constant at c, where we assume $c < a$. Following Cournot, suppose that the firms choose their quantities simultaneously.

Resolution:

Exercise 7. (item 17 of the combined list) *Homogeneous Bertrand* [Exam level; difficulty 3/5]

Source. Gibbons (1992), *Game Theory for Applied Economists*, Ch. 1 "Static Games of Complete Information", §1.5 Problems (Section 1.2), Problem 1.7, pp. 49–50 [PDF 57–58]. No answer in the book. Reproduced verbatim. The pointer to Section 1.2.B (Bertrand with differentiated products) is context; the statement defines the whole homogeneous-products model.

1.7. In Section 1.2.B, we analyzed the Bertrand duopoly model with differentiated products. The case of homogeneous products yields a stark conclusion. Suppose that the quantity that consumers demand from firm i is $a - p_i$ when $p_i < p_j$, 0 when $p_i > p_j$, and $(a - p_i)/2$ when $p_i = p_j$. Suppose also that there are no fixed costs and that marginal costs are constant at c, where $c < a$. Show that if the firms choose prices simultaneously, then the unique Nash equilibrium is that both firms charge the price c.

Resolution:

Exercise 8. (item 18 of the combined list) *Market Entry* [Exam level; difficulty 4/5]

Source. Tadelis (2013), *Game Theory: An Introduction*, Ch. 6 "Mixed Strategies", Exercise 6.8 "Market Entry", p. 125 [PDF 144]. No answer in the book. Reproduced verbatim.

6.8 Market Entry: Three firms are considering entering a new market. The payoff for each firm that enters is $\frac{150}{n}$, where n is the number of firms that enter. The cost of entering is 62.

a. Find all the pure-strategy Nash equilibria.

b. Find the symmetric mixed-strategy equilibrium in which all three players enter with the same probability.

Resolution:

Exercise 9. (item 19 of the combined list) *A unique IESDS survivor is a Nash equilibrium* [Advanced; difficulty 4/5]

Source. Mas-Colell, Whinston & Green (1995), *Microeconomic Theory*, Ch. 8 "Simultaneous-Move Games", Exercises, Exercise 8.D.2^{B}, p. 263 [PDF 142, right half]. Answer in the MWG solutions manual (`Livros/`). Reproduced verbatim.

8.D.2^{B} Show that if there is a unique profile of strategies that survives iterated removal of strictly dominated strategies, this profile is a Nash equilibrium.

Resolution:

Exercise 10. (item 20 of the combined list) *Existence in 2×2 games* [Advanced; difficulty 5/5]

Source. Mas-Colell, Whinston & Green (1995), *Microeconomic Theory*, Ch. 8 "Simultaneous-Move Games", Exercises, Exercise 8.D.6^{B}, p. 264 [PDF 143, left half]. Answer in the MWG solutions manual (`Livros/`; 8.D.6 is there). Reproduced verbatim.

8.D.6^{B} Consider any two-player game of the following form (where letters indicate arbitrary payoffs):

		Player 2	
		b_1	b_2
Player 1	a_1	u, v	ℓ, m
	a_2	w, x	y, z

Show that a mixed strategy Nash equilibrium always exists in this game. [*Hint:* Define player 1's strategy to be his probability of choosing action a_1 and player 2's to be his probability of choosing b_1; then examine the best-response correspondences of the two players.]

Resolution: